

Chapter 8

GENERALIZED LINEAR MIXED MODELS (GLMMs)

8.1 INTRODUCTION

The use of random factors is not restricted to linear mixed models, the topic of Chapter 6. For many of the same reasons as seen there, we may want to incorporate random factors into nonlinear models. That is, we may wish to build a model that accommodates correlated data, or to consider the levels of a factor as selected from a population of levels in order to make inference to that population.

For example, suppose we wish to study factors affecting cost of hospitalization by taking a random sample of patient records from each of 15 teaching hospitals. The costs within a hospital almost certainly must be regarded as correlated. They will be similar because of the general costs of running the hospital, billing practices, costs of nearby, competing hospitals, and so on. Also, a goal may be to make inferences to a larger population of research hospitals. Both of these could be accommodated by incorporating random hospital effects into the model. And a potential benefit could be gained by using best prediction technology to improve the predictions for individual hospitals.

How should the random effects be incorporated? For many problems the decision between treating a factor as fixed or random is a subtle one. As illustration, if we change the example of hospitalization costs slightly and study only three hospitals, all of which are unique and whose effects

cannot easily be regarded as a random sample, we must treat them as fixed. But this would not fundamentally change the way in which they would be incorporated into the mean of the response. This line of argument suggests that random factors should be incorporated in the same manner and in the same portion of the model as the fixed factors. This is exactly the approach of Chapter 6. Our basic linear model there had mean $E[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$. We incorporated random effects by enlarging the model as $E[\mathbf{y}|\mathbf{u}] = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$. If we write a combined model matrix $\mathbf{X}^* = [\mathbf{X} \ \mathbf{Z}]$ and an enlarged “parameter” vector $\boldsymbol{\beta}^* = [\boldsymbol{\beta}' \ \mathbf{u}']'$ it is easy to see that $E[\mathbf{y}|\mathbf{u}] = \mathbf{X}^*\boldsymbol{\beta}^*$.

This suggests a straightforward extension of the generalized linear models of Chapter 5: Append the random effects in the form $\mathbf{Z}\mathbf{u}$ to the linear predictor $\mathbf{X}\boldsymbol{\beta}$. This will achieve the two main goals of incorporating correlation and allowing broader inference. However, the nonlinear nature of the model creates complications not encountered in Chapter 6.

In the remainder of the chapter we define the generalized linear mixed model (GLMM), explore the consequences of adding random factors and discuss a variety of inferential methods. The issue of prediction of random effects we leave to Chapter 9. Models in which the random effects cannot be incorporated in a linear predictor are dealt with briefly in Chapter 11.

8.2 STRUCTURE OF THE MODEL

a. Conditional distribution of $\mathbf{y}$

To specify the model we start with the conditional distribution of $\mathbf{y}$ given $\mathbf{u}$. As in (5.5) and (5.6), the response vector $\mathbf{y}$ is typically, but not necessarily, assumed to consist of conditionally independent elements, each with a distribution with density from the exponential family or similar to the exponential family:

$$\begin{aligned} y_i|\mathbf{u} &\sim \text{indep. } f_{Y_i|\mathbf{u}}(y_i|\mathbf{u}) \\ f_{Y_i|\mathbf{u}}(y_i|\mathbf{u}) &= \exp\{[y_i\gamma_i - b(\gamma_i)]/\tau^2 - c(y_i, \tau)\}. \end{aligned} \quad (8.1)$$

From (5.12) we know that the conditional mean of y_i is related to γ_i in (8.1) via the identity $\mu_i = \partial b(\gamma_i)/\partial \gamma_i$. It is a transformation of this mean that we wish to model as a linear model in both the fixed and

random factors:

$$\begin{aligned} E[y_i|\mathbf{u}] &= \mu_i \\ g(\mu_i) &= \mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}. \end{aligned} \quad (8.2)$$

As in Chapter 5, $g(\cdot)$ is a known function, called the *link function* (since it links together the conditional mean of y_i and the linear form of predictors), $\mathbf{x}'_i$ is the i th row of the model matrix for the fixed effects, and $\boldsymbol{\beta}$ is the fixed effects parameter vector. To that specification we have added $\mathbf{z}'_i$, which is the i th row of the model matrix for the random effects, and $\mathbf{u}$, the random effects vector. Note that we are using μ_i here to denote the conditional mean of y_i given $\mathbf{u}$, not the unconditional mean. To complete the specification we assign a distribution to the random effects:

$$\mathbf{u} \sim f_{\mathbf{U}}(\mathbf{u}). \quad (8.3)$$

In light of the fact that the conditional distribution of $\mathbf{y}$ given $\mathbf{u}$ is just a notational extension of the generalized linear model of Chapter 5 (i.e., μ_i represents the conditional mean rather than the marginal or unconditional mean; otherwise, all is the same), many of the relationships derived there will hold. Correspondingly, as below (5.14), we denote the conditional variance of y_i given $\mathbf{u}$ as $\tau^2 v(\mu_i)$ in order to display its dependence on the conditional mean μ_i .

8.3 CONSEQUENCES OF HAVING RANDOM EFFECTS

a. Marginal versus conditional distribution

Since the model specification in (8.1) and (8.2) is made conditional on the value of $\mathbf{u}$, we now derive aspects of the marginal distribution of $\mathbf{y}$ in order to understand what has been assumed for the observed data.

b. Mean of $\mathbf{y}$

The mean of $\mathbf{y}$ can be derived by the usual device of iterated expectation:

$$\begin{aligned} E[y_i] &= E[E[y_i|\mathbf{u}]] \\ &= E[\mu_i] \\ &= E[g^{-1}(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u})]. \end{aligned} \quad (8.4)$$

This cannot, in general, be simplified, due to the nonlinear function $g^{-1}(\cdot)$.

To illustrate for a particular $g(\cdot)$, suppose we have a log link so that $g(\mu) = \log \mu$ and $g^{-1}(x) = \exp\{x\}$. Then we have

$$\begin{aligned} E[y_i] &= E[\exp\{\mathbf{x}'_i\beta + \mathbf{z}'_i\mathbf{u}\}] \\ &= \exp\{\mathbf{x}'_i\beta\}E[\exp\{\mathbf{z}'_i\mathbf{u}\}] \\ &= \exp\{\mathbf{x}'_i\beta\}M_{\mathbf{u}}(\mathbf{z}_i), \end{aligned} \quad (8.5)$$

where $M_{\mathbf{u}}(\mathbf{z}_i)$ is the moment generating function of $\mathbf{u}$ evaluated at $\mathbf{z}_i$ (see Section S.1c).

Suppose further that $u_i \sim \mathcal{N}(0, \sigma_u^2)$ and that each row of $\mathbf{Z}$ has a single entry equal to 1 with all the rest being zero. Then $M_{\mathbf{u}}(\mathbf{z}_i) = \exp\{\sigma_u^2/2\}$ and

$$E[y_i] = \exp\{\mathbf{x}'_i\beta\} \exp\{\sigma_u^2/2\}$$

or

$$\log E[y_i] = \mathbf{x}'_i\beta + \sigma_u^2/2. \quad (8.6)$$

c. Variances

To derive the marginal variance of $\mathbf{y}$ we use formula (1.14):

$$\begin{aligned} \text{var}(y_i) &= \text{var}(E[y_i|\mathbf{u}]) + E[\text{var}(y_i|\mathbf{u})] \\ &= \text{var}(\mu_i) + E[\tau^2 v(\mu_i)] \\ &= \text{var}(g^{-1}[\mathbf{x}'_i\beta + \mathbf{z}'_i\mathbf{u}]) + E[\tau^2 v(g^{-1}[\mathbf{x}'_i\beta + \mathbf{z}'_i\mathbf{u}])], \end{aligned} \quad (8.7)$$

which again cannot be simplified appreciably without making specific assumptions about the form of $g(\cdot)$ and/or the conditional distribution of $\mathbf{y}$.

To illustrate the derivation assume, as before, that we have a log link and now further assume that the elements of $\mathbf{y}$, given $\mathbf{u}$, are independent with a Poisson distribution. Hence the conditional variance of y_i given $\mathbf{u}$ is $\tau^2 v(\mu_i) = \mu_i$. Using these facts in (8.7) gives

$$\begin{aligned} \text{var}(y_i) &= \text{var}(\mu_i) + E[\mu_i] \\ &= \text{var}(\exp\{\mathbf{x}'_i\beta + \mathbf{z}'_i\mathbf{u}\}) + E[\exp\{\mathbf{x}'_i\beta + \mathbf{z}'_i\mathbf{u}\}] \end{aligned}$$

$$\begin{aligned}
&= E[(\exp\{2(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u})\})] - [E(\exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}\})]^2 \\
&\quad + E[\exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}\}] \quad (8.8) \\
&= \exp\{2\mathbf{x}'_i\boldsymbol{\beta}\} \left(M_u(2\mathbf{z}_i) - [M_u(\mathbf{z}_i)]^2 + \exp\{-\mathbf{x}'_i\boldsymbol{\beta}\} M_u(\mathbf{z}_i) \right).
\end{aligned}$$

If we make the further assumption that $u_i \sim \mathcal{N}(0, \sigma_u^2)$ and that each row of $\mathbf{Z}$ has a single entry equal to 1 with all the rest being zero, then

$$\begin{aligned}
\text{var}(y_i) &= \exp\{2\mathbf{x}'_i\boldsymbol{\beta}\} \left(\exp\{2\sigma_u^2\} - \exp\{\sigma_u^2\} \right) + \exp\{\mathbf{x}'_i\boldsymbol{\beta}\} \exp\{\sigma_u^2/2\} \\
&= \exp\{\mathbf{x}'_i\boldsymbol{\beta} + \sigma_u^2/2\} \left(\exp\{\mathbf{x}'_i\boldsymbol{\beta}\} \left[\exp\{3\sigma_u^2/2\} - \exp\{\sigma_u^2/2\} \right] + 1 \right) \\
&= E[y_i] \left(\exp\{\mathbf{x}'_i\boldsymbol{\beta}\} \left[\exp\{3\sigma_u^2/2\} - \exp\{\sigma_u^2/2\} \right] + 1 \right). \quad (8.9)
\end{aligned}$$

Since the term in parentheses in (8.9) is greater than 1, we see that the variance is larger than the mean. Therefore, although the conditional distribution of y_i given $\mathbf{u}$ is Poisson, the marginal distribution cannot be. In fact, under these assumptions, it will always be overdispersed (see Section 2.6b-ii) compared to the Poisson distribution. In this sense we can think of random effects as a way to model or attribute overdispersion to a particular source.

d. Covariances and correlations

As noted before, the use of random effects introduces a correlation among observations which have any random effect in common. The same is true for generalized linear mixed models. Assuming conditional independence of the elements of $\mathbf{y}$ and using (1.16), we have

$$\begin{aligned}
\text{cov}(y_i, y_j) &= \text{cov}(E[y_i|\mathbf{u}], E[y_j|\mathbf{u}]) + E[\text{cov}(y_i, y_j|\mathbf{u})] \\
&= \text{cov}(\mu_i, \mu_j) + E[0] \\
&= \text{cov}(g^{-1}[\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}], g^{-1}[\mathbf{x}'_j\boldsymbol{\beta} + \mathbf{z}'_j\mathbf{u}]). \quad (8.10)
\end{aligned}$$

If we have a log link, this can be evaluated as

$$\begin{aligned}
\text{cov}(y_i, y_j) &= \text{cov}(\exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}\}, \exp\{\mathbf{x}'_j\boldsymbol{\beta} + \mathbf{z}'_j\mathbf{u}\}) \\
&= \exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{x}'_j\boldsymbol{\beta}\} \text{cov}(\exp\{\mathbf{z}'_i\mathbf{u}\}, \exp\{\mathbf{z}'_j\mathbf{u}\}) \quad (8.11) \\
&= \exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{x}'_j\boldsymbol{\beta}\} [M_u(\mathbf{z}_i + \mathbf{z}_j) - M_u(\mathbf{z}_i)M_u(\mathbf{z}_j)].
\end{aligned}$$

Again we make further assumptions, namely that $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}\sigma_u^2)$ and that each row of $\mathbf{Z}$ has a single entry equal to 1 with all the rest being zero. Then

$$\text{cov}(y_i, y_j) = \exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{x}'_j\boldsymbol{\beta}\} \left[\exp\{\sigma_u^2\}(\exp\{\mathbf{z}'_i\mathbf{z}_j\sigma_u^2\} - 1) \right] \quad (8.12)$$

which is equal to zero if $\mathbf{z}'_i\mathbf{z}_j = 0$ (i.e., if the two observations do not share a random effect) and is positive otherwise (in which case $\mathbf{z}'_i\mathbf{z}_j = 1$).

From (8.12) and (8.9), when $\mathbf{z}'_i\mathbf{z}_j = 1$, we can calculate the correlation (after canceling $\exp\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{x}'_j\boldsymbol{\beta}\}$ in the numerator and denominator) as:

$$\begin{aligned} \text{corr}(y_i, y_j) &= \frac{e^{2\sigma_u^2} - e^{\sigma_u^2}}{\sqrt{\left(e^{2\sigma_u^2} - e^{\sigma_u^2} + e^{-\mathbf{x}'_i\boldsymbol{\beta} + \sigma_u^2/2}\right) \left(e^{2\sigma_u^2} - e^{\sigma_u^2} + e^{-\mathbf{x}'_j\boldsymbol{\beta} + \sigma_u^2/2}\right)}} \\ &= \frac{1}{\sqrt{\left(1 + \eta e^{-\mathbf{x}'_i\boldsymbol{\beta}}\right) \left(1 + \eta e^{-\mathbf{x}'_j\boldsymbol{\beta}}\right)}}, \end{aligned} \quad (8.13)$$

where η is given by $1/(e^{3\sigma_u^2/2} - e^{\sigma_u^2/2})$.

8.4 ESTIMATION BY MAXIMUM LIKELIHOOD

a. Likelihood

From (8.1), (8.2), and (8.3) it is straightforward to write down a formula for the likelihood:

$$L = \int \prod_i f_{Y_i|\mathbf{u}}(y_i|\mathbf{u}) f_{\mathbf{U}}(\mathbf{u}) d\mathbf{u}, \quad (8.14)$$

where, as before, the integration is over the q -dimensional distribution of $\mathbf{u}$.

As an example, consider modeling data in correlated clusters thought to come from a Poisson distribution. An example of such a situation is described in Diggle et al. (1994) in which they consider the analysis of the number of epileptic seizures in patients on a drug or placebo. In this context, the clusters would be repeated measurements taken

on the same patients. Let y_{ij} denote the j th count taken on the i th cluster. We might therefore create a model as:

$$\begin{aligned} y_{ij} | \mathbf{u} &\sim \text{indep. Poisson}(\mu_{ij}); \quad i = 1, 2, \dots, m; j = 1, 2, \dots, n_i; \\ \log \mu_{ij} &= \mathbf{x}'_{ij} \boldsymbol{\beta} + u_i \\ u_i &\sim \text{i.i.d. } \mathcal{N}(0, \sigma_u^2). \end{aligned} \quad (8.15)$$

This uses a log link and a normal distribution for the random cluster (patient) effects. The normal distribution assumption for the random effects is viable since the log link carries the range of the parameter space for μ_{ij} into the entire real line. The random effects u_i are shared among observations within the same cluster and hence those observations are being modeled as correlated.

The log likelihood can be simplified as follows (see E 8.3)

$$\begin{aligned} l &= \log \left(\prod_{i=1}^m \int_{-\infty}^{\infty} \prod_{j=1}^{n_i} \frac{\mu_{ij}^{y_{ij}} e^{-\mu_{ij}}}{y_{ij}!} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} u_i^2} du_i \right) \\ &= \mathbf{y}' \mathbf{X} \boldsymbol{\beta} - \sum_{i,j} \log y_{ij}! \\ &\quad + \sum_i \log \int_{-\infty}^{\infty} \exp \left\{ y_i u_i - \sum_j x'_{ij} \boldsymbol{\beta} + u_i \right\} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} u_i^2} du_i. \end{aligned} \quad (8.16)$$

Unfortunately, (8.16) cannot be simplified further or evaluated in closed form and hence maximizing values cannot be expressed in closed form either.

In the simplest cases, numerical integration for calculating the likelihood is straightforward and hence numerical maximization of the likelihood is not too difficult. For example, for (8.15), as seen in (8.16), the log likelihood is the sum of independent contributions from each cluster, each of which involves just a single-dimensional integral. This integral can be evaluated accurately using standard quadrature techniques, for example, Gauss-Hermite quadrature (see Chapter 10).

This “brute force” approach to maximum likelihood works relatively well in simple situations: a single random effect, two or perhaps three nested random effects, and random effects which come in clusters (e.g., longitudinal data with subjects having random intercepts and slopes). However, for more complicated structures (e.g., crossed random factors) it fails.

b. Likelihood equations

– i. For the fixed effects parameters

Even though the likelihood equations are numerically difficult, we can write them in a simpler form. From (8.14)

$$l = \log \int f_{Y|U}(y|u) f_U(u) du = \log f_Y(y), \quad (8.17)$$

so that

$$\begin{aligned} \frac{\partial l}{\partial \beta} &= \frac{\partial}{\partial \beta} \int f_{Y|U}(y|u) f_U(u) du / f_Y(y) \\ &= \int \left[\frac{\partial}{\partial \beta} f_{Y|U}(y|u) \right] f_U(u) du / f_Y(y), \end{aligned} \quad (8.18)$$

since $f_U(u)$ does not involve β . Noting that

$$\begin{aligned} \frac{\partial}{\partial \beta} f_{Y|U}(y|u) &= \left(\frac{1}{f_{Y|U}(y|u)} \frac{\partial f_{Y|U}(y|u)}{\partial \beta} \right) f_{Y|U}(y|u) \\ &= \frac{\partial \log f_{Y|U}(y|u)}{\partial \beta} f_{Y|U}(y|u) \end{aligned} \quad (8.19)$$

we can rewrite (8.18) as

$$\begin{aligned} \frac{\partial l}{\partial \beta} &= \int \frac{\partial \log f_{Y|U}(y|u)}{\partial \beta} f_{Y|U}(y|u) f_U(u) du / f_Y(y) \\ &= \int \frac{\partial \log f_{Y|U}(y|u)}{\partial \beta} f_{U|Y}(u|y) du. \end{aligned} \quad (8.20)$$

Using (5.18), which gives the derivative of the log likelihood for a GLM, in (8.20) gives

$$\begin{aligned} \frac{\partial l}{\partial \beta} &= \int \mathbf{X}' \mathbf{W}^* (\mathbf{y} - \boldsymbol{\mu}) f_{U|Y}(u|y) du \\ &= \mathbf{X}' \mathbf{E}[\mathbf{W}^* | \mathbf{y}] - \mathbf{X}' \mathbf{E}[\mathbf{W}^* \boldsymbol{\mu} | \mathbf{y}], \end{aligned} \quad (8.21)$$

where $\mathbf{W}^* = \left\{ a(\phi) v(\mu_i) g_\mu(\mu_i) \right\}^{-1}$.

The likelihood equation for β is therefore

$$\mathbf{X}' \mathbf{E}[\mathbf{W}^* | \mathbf{y}] = \mathbf{X}' \mathbf{E}[\mathbf{W}^* \boldsymbol{\mu} | \mathbf{y}], \quad (8.22)$$

which is similar to (5.19), the difference being that $\mathbf{W}^*$ and $\mathbf{W}^*\boldsymbol{\mu}$ are replaced by their conditional expected values given $\mathbf{y}$.

In cases like the Poisson example of (8.15), $\mathbf{W}^* = \mathbf{I}$ and the equations simplify to

$$\mathbf{X}'\mathbf{y} = \mathbf{X}'\mathbf{E}[\boldsymbol{\mu}|\mathbf{y}]. \quad (8.23)$$

Computing issues related to solving these equations are described in Chapter 10.

– ii. *For the random effects parameters*

A result similar to (8.20) can be derived for the ML equations for the parameters in the distribution of $f_{\mathbf{U}}(\mathbf{u})$. Let $\boldsymbol{\varphi}$ denote those parameters so that

$$\begin{aligned} \frac{\partial l}{\partial \boldsymbol{\varphi}} &= \int \frac{\partial \log f_{\mathbf{U}}(\mathbf{u})}{\partial \boldsymbol{\varphi}} f_{\mathbf{U}|\mathbf{Y}}(\mathbf{u}|\mathbf{y}) d\mathbf{u} \\ &= \mathbf{E} \left[\frac{\partial \log f_{\mathbf{U}}(\mathbf{u})}{\partial \boldsymbol{\varphi}} \middle| \mathbf{y} \right]. \end{aligned} \quad (8.24)$$

Further simplifications are not possible without specifying a form for the random effects distribution.

8.5 MARGINAL VERSUS CONDITIONAL MODELS

Instead of starting from the conditional specification as in (8.1), (8.2), and (8.3), we might directly hypothesize a model for the mean of $\mathbf{y}$. As an example, suppose y_{ij} is equal to 1 if the j th child of woman i is born prematurely and is zero otherwise and assume we have a single predictor x_{ij} = number of drinks of alcohol per day. The marginal approach would model the marginal mean of y_{ij} directly by, for example, assuming that a logistic regression model, as in (3.109), fits the data:

$$\text{logit}(\mathbf{E}[y_{ij}]) = \text{logit}(P\{y_{ij} = 1\}) = \alpha + \beta x_{ij}.$$

In words, the model would be for logit of the probability of premature birth, averaged over a population of women. Of course, if the model was for correlated data, we would not be able to assume the observations were independent.

On the other hand, our typical conditional approach corresponds to hypothesizing the existence of a random factor for women and specifying a conditional model such as

$$\text{logit}(\mathbf{E}[y_{ij}|\mathbf{u}]) = \alpha + \beta x_{ij} + u_i,$$

where u_i represents the random woman effect. This corresponds to modeling the conditional probability of a premature birth for each woman.

From a probabilistic perspective, we can calculate the marginal distribution of $\mathbf{y}$ (at least conceptually—it might be computationally difficult) from the distribution of $\mathbf{u}$ and the conditional distribution of $\mathbf{y}|\mathbf{u}$. It is not possible to recover the marginal of $\mathbf{u}$ and the conditional distribution of $\mathbf{y}|\mathbf{u}$ from the marginal distribution of $\mathbf{y}$. This would seem to favor the conditional specification of the model.

However, in some cases, the marginal distribution (or perhaps only the marginal mean) may be adequate for answering questions of interest. For example, in the alcohol consumption example, a natural question of interest is how much could the incidence of premature birth be reduced by lowering, on average, women's alcohol consumption. In such cases, the potentially difficult problem of specifying the conditional distribution of $\mathbf{y}|\mathbf{u}$ and the marginal distribution of $\mathbf{u}$ can be avoided. This is an advantage of marginal modeling and the basis of the generalized estimating equations approach, which is described in Section 8.6a.

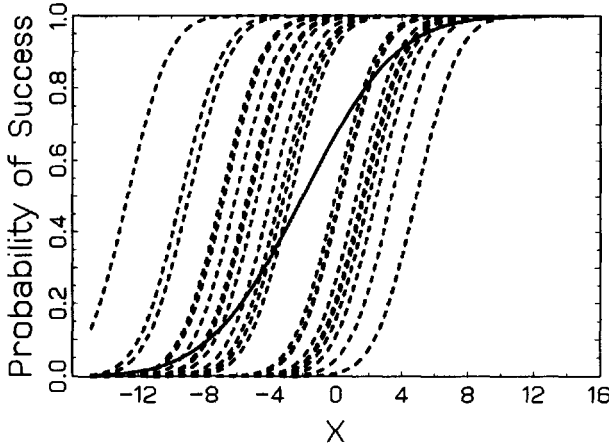
Distinguishing conditional from marginal models is straightforward probabilistically, but it is often difficult in practice. For example, a researcher might be interested in “the influence of alcohol consumption on premature birth”, which would not specify which type of model to build. In our experience, researchers often think about building models in a mechanistic way, which seems more compatible with the conditional approach. Again considering the premature birth example, a researcher might think about the influence of alcohol consumption by trying to understand how alcohol influenced individual women's physiology.

The distinction between conditional and marginal models is an important one to keep in mind in practice. The reason is perhaps easiest to see in a binary data, probit-normal model:

$$\begin{aligned} E[y_i] = P\{y_i = 1\} &= \Phi(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}), \\ \mathbf{u} &\sim \mathcal{N}(\mathbf{0}, \mathbf{D}). \end{aligned} \tag{8.25}$$

It is not hard to show, as we do later in (8.52), that if the conditional

Figure 8.1: Probability of success versus a predictor for the marginal and conditional versions of a probit-normal model. Solid line, marginal model; Dashed lines, realizations of the conditional model.



mean is given as in (8.25), then the unconditional mean is

$$E[y_i] = \Phi \left(\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sqrt{\mathbf{z}_i' \mathbf{D} \mathbf{z}_i + 1}} \right) \equiv \Phi(\mathbf{x}_i' \boldsymbol{\beta}^*) \quad (8.26)$$

so that $\boldsymbol{\beta}^* = \boldsymbol{\beta} / \sqrt{\mathbf{z}_i' \mathbf{D} \mathbf{z}_i + 1}$. Hence $\boldsymbol{\beta}$ represents the magnitude of the effect of the predictors on the conditional distribution, while $\boldsymbol{\beta}^*$ represents the magnitude of the effect on the marginal distribution.

Clearly, $\boldsymbol{\beta}$ is always larger than $\boldsymbol{\beta}^*$ in absolute value. Why this is so is perhaps easiest to understand graphically, as shown in Figure 8.1. Each of the realized values of the conditional model [i.e., equation (8.25) plotted for various realized u_i], shown by the dashed lines in the figure, have a large value of $\boldsymbol{\beta}$. However, the variance of the random effect is quite large and when all the curves are averaged, the resulting unconditional mean has a much smaller slope and much smaller value for $\boldsymbol{\beta}^*$.

There are other advantages to the conditional approach. For example, if two studies are performed in different populations with different variances the marginal models will be different even though the conditional models are the same. Returning to the alcohol consumption example, suppose a small, preliminary study is conducted with a pur-

posefully homogeneous study population (and hence a small variance for the subject random effect). Later, a larger scale study is conducted on a wider scale with a more heterogeneous study population. Even if the effect on every person in both studies is the same, the marginal models will differ because of the different variances.

8.6 OTHER METHODS OF ESTIMATION

The difficulty in evaluating the likelihood for models such as (8.15) and the fact that numerical maximization must be resorted to has led to both alternative approaches and to a body of research for effective ways to compute and maximize the likelihoods. The latter topic is treated in Chapter 10; here we introduce some of the alternative methods of estimation.

a. Generalized estimating equations

The generalized estimating equations (GEEs) approach begins by positing a marginal generalized linear model for the mean of $\mathbf{y}$ as a function of the predictors. For example, for binary data we might hypothesize a logistic regression for the mean:

$$\text{logit}(E[\mathbf{y}]) = \mathbf{X}\beta.$$

If we used a working assumption of independence (as in Section 7.8) of all the elements of $\mathbf{y}$, the ML estimating equations for β would be, from E 5.9,

$$\mathbf{X}'\mathbf{y} = \mathbf{X}'E[\mathbf{y}]. \quad (8.27)$$

These are *unbiased estimating equations*, meaning that the difference between the right-hand side and the left-hand side is zero, namely $E(\mathbf{X}'\mathbf{y} - \mathbf{X}'E[\mathbf{y}]) = \mathbf{0}$. It is not surprising and is true under regularity conditions (Heyde, 1997, Sec. 12.2) that solutions to unbiased estimating equations give consistent estimators.

Operationally, this estimator could be calculated by pretending that all the data were independent and conducting a standard logistic regression analysis. Just as described in Section 7.8, this is often a nearly fully efficient estimator but the standard errors, variance estimates, tests and confidence intervals would often be highly misleading. Again, as in Section 7.8, this can be dealt with by using the *estimator* that naively assumes independence but properly calculating its (large-sample) variance.

For longitudinal data with m subjects and with $\mathbf{y}_i$ denoting the data for the i th subject, we have

$$\mathbf{y} = \left\{ \mathbf{y}_i \right\}_{i=1}^m \quad \text{and} \quad \mathbf{X} = \left\{ \mathbf{X}_i \right\}_{i=1}^m$$

and the estimating equation, (8.27), for binary data becomes

$$\sum \mathbf{X}_i \mathbf{y}_i = \sum \mathbf{X}_i E[\mathbf{y}_i]. \quad (8.28)$$

From this it can be shown that (Heyde, 1997, Secs. 4.2 and 12.4) the large-sample variance of $\hat{\beta}$ is

$$\begin{aligned} \text{var}_{\infty}(\hat{\beta}) &= (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \text{var}(\mathbf{y}) \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \\ &= \left(\sum \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\sum \mathbf{X}_i' \text{var}(\mathbf{y}_i) \mathbf{X}_i \right) \left(\sum \mathbf{X}_i' \mathbf{X}_i \right)^{-1}, \end{aligned} \quad (8.29)$$

upon the assumption of independence among the $\mathbf{y}_i$. This can be consistently estimated as $m \rightarrow \infty$ with

$$\widehat{\text{var}}_{\infty}(\hat{\beta}) = \left(\sum \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\sum \mathbf{X}_i' (\mathbf{y}_i - \hat{E}[\mathbf{y}_i]) (\mathbf{y}_i - \hat{E}[\mathbf{y}_i])' \mathbf{X}_i \right) \left(\sum \mathbf{X}_i' \mathbf{X}_i \right)^{-1}, \quad (8.30)$$

where $\hat{E}[\mathbf{y}_i] = 1/(1 + \exp\{-\mathbf{X}_i \hat{\beta}\})$.

The working assumption of independence may lead to inefficient estimators (Fitzmaurice, 1995) and other working variance-covariance structures can be entertained. In that case, for (8.28) we have

$$\sum \mathbf{X}_i \mathbf{W}_i \mathbf{y}_i = \sum \mathbf{X}_i \mathbf{W}_i E[\mathbf{y}_i], \quad (8.31)$$

where $\mathbf{W}_i^{-1} = \text{var}_W(\mathbf{y}_i)$ is the working variance for $\mathbf{y}_i$. This engenders corresponding changes in (8.29) and (8.30). See Diggle et al. (1994, Secs. 8.2.3 and 8.4.2) for more details.

b. Penalized quasi-likelihood

For the generalized linear models (GLMs) of Chapter 5, the use of working variates, as in (5.4) and (5.28), and the principle of quasi-likelihood (Section 5.6) are highly useful concepts. Quasi-likelihood is attractive because of its ability to generate highly efficient estimators without making precise distributional assumptions. Working variates form the basis of efficient computing algorithms for both maximum likelihood and maximum quasi-likelihood. A natural question is whether they can be adapted for use in GLMMS.

Working variates for GLMs begin with a Taylor expansion of the link function around the mean of y_i :

$$t_i \equiv \mathbf{x}_i' \boldsymbol{\beta} + g_\mu(\mu_i)(y_i - \mu_i).$$

The working variate thus follows a linear model and can be used to form a provisional estimate of $\boldsymbol{\beta}$. This local approximation is repeated at each update of an iterative algorithm.

The direct analog of working variates for the GLMM specification in (8.1) and (8.2) would be an expansion around the conditional mean of y_i :

$$t_i \equiv \mathbf{x}_i' \boldsymbol{\beta} + \mathbf{z}_i' \mathbf{u} + g_\mu(\mu_i)(y_i - \mu_i)$$

or

$$\mathbf{t} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu}), \quad (8.32)$$

where $\boldsymbol{\Delta} = \left\{ \frac{d}{d\mu} g_\mu(\mu_i) \right\}$. To derive a local approximation, the next step would be to calculate the variance of $\mathbf{t}$. But this approach quickly becomes complicated since $\boldsymbol{\Delta}$ (through its dependence on $\boldsymbol{\mu}$) and $\boldsymbol{\mu} = E[\mathbf{y}|\mathbf{u}]$ itself are random functions of $\mathbf{u}$ and their variances are not easily calculated.

A possible simplification is to set $\mathbf{u}$ in $\boldsymbol{\Delta}$ equal to its mean, $\mathbf{0}$, simplifying (8.32) to be

$$\mathbf{t} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\Delta}^*(\mathbf{y} - \boldsymbol{\mu}), \quad (8.33)$$

where $\boldsymbol{\Delta}^* = \left\{ \frac{d}{d\mu} g_\mu[g^{-1}(\mathbf{x}_i' \boldsymbol{\beta})] \right\}$.

Under this simplification

$$\begin{aligned} \text{var}(\mathbf{t}) &= \mathbf{Z}\mathbf{D}\mathbf{Z}' + \boldsymbol{\Delta}^* \text{var}(\mathbf{y} - \boldsymbol{\mu}) \boldsymbol{\Delta}^* \\ &\equiv \mathbf{Z}\mathbf{D}\mathbf{Z}' + \mathbf{R}. \end{aligned} \quad (8.34)$$

That is, the working variate $\mathbf{t}$ approximately follows a linear mixed model (LMM) as in Chapter 6. This suggests an iterative algorithm (Schall, 1991) in which an LMM is fitted to get estimates of $\boldsymbol{\beta}$ and $\mathbf{u}$. These are then used to recalculate the working variate, and so on.

A completely different justification of this approach is via what is called *penalized quasi-likelihood* (PQL). Recall that quasi-likelihood

does not specify a distribution, only the mean-to-variance relationship. This is not a sufficient basis on which to estimate the variance-covariance structure. One suggestion (Green and Silverman, 1994) to remedy this defect is to add a penalty function to the quasi-likelihood of the form $\frac{1}{2}\mathbf{u}'\mathbf{D}^{-1}\mathbf{u}$, that is

$$\text{PQL} = \sum Q_i - \frac{1}{2}\mathbf{u}'\mathbf{D}^{-1}\mathbf{u}, \quad (8.35)$$

where Q_i is defined in (5.50).

The maximum quasi-likelihood equations would come from differentiating (8.35) with respect to β and $\mathbf{u}$ and would be [compare (5.53)]

$$\frac{1}{\tau^2}\mathbf{X}'\mathbf{W}\Delta(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$$

and

$$\frac{1}{\tau^2}\mathbf{Z}'\mathbf{W}\Delta(\mathbf{y} - \boldsymbol{\mu}) - \mathbf{D}^{-1}\mathbf{u} = \mathbf{0}. \quad (8.36)$$

These lead (Breslow and Clayton, 1993) to a computational algorithm similar to that of Schall(1991). Yet another justification for this approach is via Laplace approximations (see Chapter 10 and Wolfinger, 1994).

Despite the number of ways in which basically the same approach has been justified, it has not been found to work well in practice, especially for binary data in small clusters (Breslow and Clayton, 1993; Breslow and Lin, 1995; Lin and Breslow, 1996). We therefore recommend that unmodified penalized quasi-likelihood not be used in practice. More detail is given in Chapter 10.

c. Conditional likelihood

An approach very different in nature to integrating random effects out of the distribution and working with the marginal distribution is to consider a conditional likelihood and construct conditional estimators and tests as in Section 3.8e. In the conditional approach we start with the conditional distribution of the data given the random effects, but instead of hypothesizing a distribution for these effects and integrating them out, they are treated as fixed parameters. The sufficient statistics for them are derived and the conditional distribution given the sufficient statistics (which, by definition, is free of these effects) is used for inferential purposes.

A classic application of the conditional approach is to the case of matched pairs binary data. For example, suppose we wish to ask whether cancer patients get more effective treatment in major cancer centers than in community hospitals. We cannot compare remission rates directly since patient populations might be drastically different. For example, major cancer centers might appear to provide poorer treatment merely because they treat the most difficult cases.

A possible solution is to employ a matched pairs design: A patient from a cancer center is matched with a patient from a community hospital on the basis of treatment date, type of treatment received, and patient's age. Suppose that the response variable is whether or not there is a sizable shrinkage in tumor size within 90 days and let $y_{ij} = 1$ for shrinkage and 0 otherwise. Here i indexes pairs $i = 1, 2, \dots, n$ and j indexes type of hospital (with $j = 1$ representing a cancer center and $j = 2$ representing a community hospital). Also, let x_{ij} be 0 when $j = 1$ and 1 when $j = 2$.

A possible model for y_{ij} is:

$$y_{ij}|u_i \sim \text{indep. Bernoulli}[\pi(x_{ij})]$$

$$\text{logit}[\pi(x_{ij})] = \alpha + u_i + \beta x_{ij}$$

Primary interest focuses on β , which represents the log odds of tumor shrinkage for community hospitals as compared to cancer centers, which is assumed constant within each pair. The u_i represent the pair-to-pair differences in the probability of tumor shrinkage.

What happens if we treat the u_i as fixed effects and estimate them, along with β ? Maximum likelihood gives (see E 8.6).

$$\hat{\beta} = 2 \log \frac{N_{01}}{N_{10}}, \tag{8.37}$$

where N_{10} is the number of pairs with $y_{i1} = 1$ and $y_{i2} = 0$ and where N_{01} is the number of pairs with $y_{i1} = 0$ and $y_{i2} = 1$. This is perhaps easiest to visualize in a 2×2 format as in Table 8.1.

Table 8.1: Fate of Matched Pairs

Cancer Center	Community Hospital	
	Success	Failure
Success	N_{11}	N_{10}
Failure	N_{01}	N_{00}

It is not hard to show that this ML estimator is twice what it “should” be in the sense that it converges to 2β (see E 8.7).

What is the remedy? A commonly-used approach is that of conditional likelihood, in which we derive the sufficient statistics for the u_i and work with the conditional distribution given those sufficient statistics.

We follow the development in (3.134) through (3.138). If interest focuses on β , then we will want to base inferences on the conditional distribution of $T = \sum_{i,j} y_{ij}x_{ij}$ given $S_1 = y_{1\cdot}, S_2 = y_{2\cdot}, \dots, S_m = y_{m\cdot}$. In our example,

$$\begin{aligned} T &= \sum_{i,j} y_{ij}x_{ij} \\ &= \sum_i y_{i2} \\ &= y_{\cdot 2} \\ &= \text{number of successes in community hospitals,} \end{aligned} \quad (8.38)$$

so we want the conditional distribution of the total number of successes in the community hospitals conditional on the number of successes in each pair. From Table 8.1, $T = N_{11} + N_{01}$. Now N_{11} is just the number of pairs that have two successes, so conditional on S_i , it is known and fixed. We therefore focus on the conditional distribution of N_{01} given the S_i . We build it up in two steps. First consider a pair for which $S_i = 1$.

If $S_i = 1$, there are two possibilities: $\{y_{i1} = 0, y_{i2} = 1\}$ or $\{y_{i1} = 1, y_{i2} = 0\}$. The conditional probability of the first event is

$$\begin{aligned} P\{y_{i1} = 1, y_{i2} = 0 | S_i = 1\} &= \frac{P\{y_{i1} = 1, y_{i2} = 0\}}{P\{y_{i1} = 0, y_{i2} = 1\}} + P\{y_{i1} = 1, y_{i2} = 0\} \\ &= \frac{\left(1 - \frac{1}{1+e^{-(\alpha+u_i)}}\right) \frac{1}{1+e^{-(\alpha+u_i+\beta)}}}{\left(1 - \frac{1}{1+e^{-(\alpha+u_i)}}\right) \frac{1}{1+e^{-(\alpha+u_i+\beta)}} + \frac{1}{1+e^{-(\alpha+u_i)}} \left(1 - \frac{1}{1+e^{-(\alpha+u_i+\beta)}}\right)} \\ &= \frac{e^{-(\alpha+u_i+\beta)}}{e^{-(\alpha+u_i)} + e^{-(\alpha+u_i+\beta)}} \\ &= \frac{1}{1 + e^{-\beta}}. \end{aligned} \quad (8.39)$$

So the conditional distribution of y_{i2} given $S_i = 1$ is Bernoulli with a probability of success having a logit of β .

Writing T as

$$\begin{aligned} T &= N_{11} + N_{01} \\ &= \sum_{S_i=2} 1 + \sum_{S_i=1} I_{\{y_{i2}=1\}}, \end{aligned} \quad (8.40)$$

shows that the conditional distribution of $N_{01} = T - N_{11}$ given the S_i is the sum of independent Bernoullis, each with conditional probability of success of $(1 + e^{-\beta})^{-1}$, that is,

$$N_{01}|S \sim \text{binomial}\left(N_{01} + N_{10}, \frac{1}{1 + e^{-\beta}}\right). \quad (8.41)$$

It is therefore straightforward to show that the maximizing value of the conditional distribution (the conditional MLE) is

$$\hat{\beta} = \log \frac{N_{01}}{N_{10}}, \quad (8.42)$$

which is a consistent estimator of β as $N_{11} + N_{01}$ increases.

Also, under $H_0: \beta = 0$, the distribution is $\text{binomial}(N_{11} + N_{01}, \frac{1}{2})$, from which exact tests or p -values can easily be derived. To do so, we use as our test statistic N_{01} , the number of successes in the community hospitals out of the pairs for which $S_i = 1$. If we were testing against the alternative that $H_A: \beta > 0$, we would reject for large values of N_{01} . Therefore the p -value for the one-tailed test would be:

$$p\text{-value} = P\{X \geq N_{01}\}, \quad (8.43)$$

where $X \sim \text{binomial}(N_{01} + N_{10}, \frac{1}{2})$.

As a numerical illustration, consider the hypothetical data of Table 8.2.

Table 8.2: Matched Pairs Data

Cancer Center	Community Hospital		Total
	Success	Failure	
Success	501	157	658
Failure	146	132	278
Total	647	289	936

The conditional analysis discards the $501 + 132 = 633$ responses for which the response in the cancer center and community hospital are

the same and bases the analysis on the 303 remaining. The p -value for the one-tailed test is

$$\begin{aligned} p\text{-value} &= P\{X \geq 157\} \\ \text{where } X &\sim \text{binomial}(303, 1/2), \end{aligned} \quad (8.44)$$

which is approximately 0.283. A usual convention is to multiply the one-sided p -value by 2 to get the two-tailed p -value, so the answer is $2(0.283) = 0.566$, suggesting no difference between cancer centers and community hospitals.

By its nature the conditional approach has three potentially serious drawbacks. First, because it treats the random effects as unknown parameters to be conditioned away, it is incapable of making inference to quantities involving the random effects: for example, their variances or predicted values. Second, it discards information that might be available by making only weak assumptions about the form of the distribution from which the random effects are chosen. Third, it removes any information that would be gained by comparing across levels of the random effects. In some situations, virtually *all* the information of interest is garnered from comparisons across levels of a random effect. Hence use of a conditional approach would be disastrous in such a context: It would eliminate all the information of interest due to the extreme manner in which the effects are handled.

Basically, the conditional approach is effective and attractive when interest centers almost exclusively on effects that can be measured *within* levels of a random factor. When extensive information exists across levels of a random factor or when interest focuses on the random factor itself, the conditional approach cannot be used.

d. Simpler models

To avoid the computational difficulties of GLMMs, other models have been considered, mostly on the basis of computational convenience. For example, the beta-binomial model of Section 2.6b has long been used for modeling correlated binary data. The most basic use of the beta-binomial is for a context in which there are m groups, and within the i th group we have n_i observations, $y_{ij} \sim \text{binomial}(k_{ij}, p_{ij})$, with j running from 1 to n_i . This is conditional on the values of the p_{ij} . To complete the specification we assume that $p_{ij} \sim \text{beta}(\alpha_i, \beta_i)$. Given these distributional assumptions it is straightforward to show that the

likelihood is given by

$$L = \prod_{i=1}^m \prod_{j=1}^{n_i} \frac{B(\alpha_i + y_{ij}, \beta_i + k_{ij} - y_{ij})}{B(\alpha_i, \beta_i)}, \quad (8.45)$$

where $B(\alpha, \beta)$ represents the beta function.

Working with the log likelihood, some simplifications occur as in (2.79). However, the likelihood still cannot be maximized in closed form, and numerical maximization must be resorted to. Likelihood ratio tests can be performed to test for dispersion or to compare the means of the various groups.

For data having Poisson distributions, a natural distribution to incorporate correlation is the gamma distribution. As with the beta-binomial model, we consider a situation with m groups, and within group i we have n_i observations, $y_{ij} \sim \text{Poisson}(\lambda_{ij})$ with j running from 1 to n_i . This is conditional on the values of the λ_{ij} . To complete the specification we assume that $\lambda_{ij} \sim \text{gamma}(r_i, \beta_i)$. This allows for easy integration over the distribution of the parameters across groups, so that the likelihood takes a simple closed form.

However, like the beta-binomial model, it is limited in its application. It cannot handle models for the fixed effects (e.g., a regression situation), it cannot separate sources of variation in a crossed design, and it generally does not have the flexibility to tackle a wide variety of practical problems. Generalizations to handle more complicated covariate patterns and more complicated random effects structures have been considered by Lee and Nelder (1996), at the cost of additional computational complexity.

8.7 TESTS OF HYPOTHESES

The usual large-sample tools (see Chapter 2 and Sections S.4 and S.5) are about the only techniques currently available for statistical inference.

a. Likelihood ratio tests

The likelihood ratio test for nested models can be performed in the usual way by comparing $-2\log \Lambda$ to a chi-square distribution. Testing whether a variance component is zero leads to the same boundary-of-the-parameter-space problem noted before in Chapter 2 [see (2.88)]. In

the simple case where we are testing the null hypothesis that a single variance component is equal to zero, the large-sample distribution is a 50/50 mixture of the constant 0 and a χ_1^2 distribution. The critical values are thus given by (see E 8.4) $\chi_{1,1-2\alpha}^2$ for an α -level test.

Since the likelihood cannot, in general, be evaluated analytically the same is true of the likelihood ratio test statistic. It can be calculated only numerically for a given data set. In many cases it is a challenge even to perform the numerical maximization and calculation.

b. Asymptotic variances

Again, with the difficulty of calculating the likelihood, even large-sample variances and standard errors can be a computational burden. Numerical methods must be resorted to to calculate even the observed Fisher information (i.e., the negative of the second derivative matrix of the log likelihood).

c. Wald tests

For large samples, when construction of the observed or expected information is possible, Wald tests can be formed by utilizing the large-sample normality of estimators. This can be for an individual parameter:

$$\frac{\hat{\beta}_i - \beta_{i,0}}{\sqrt{\widehat{\text{var}}_\infty(\hat{\beta}_i)}} \sim \mathcal{N}(0, 1) \quad (8.46)$$

or for a set of linear combinations of the parameters,

$$\mathbf{K}'\hat{\boldsymbol{\beta}} - \mathbf{K}'\boldsymbol{\beta}_0 \sim \mathcal{N}(0, \mathbf{K}'\mathbf{I}^{-1}\mathbf{K}), \quad (8.47)$$

where $\mathbf{I}$ represents the observed or expected information.

d. Score tests

For testing the presence of a single random effect or multiple random effects, score tests have also been proposed (Commenges et al., 1994; Jacqmin-Gadda and Commenges, 1995; Lin, 1997; Commenges and Jacqmin-Gadda, 1997). These have the advantage of not requiring the maximum likelihood estimators under the GLMM. However, they often have less power than the tests based on the random effects models.

8.8 ILLUSTRATION: CHESTNUT LEAF BLIGHT

The American chestnut tree used to be a predominant hardwood in the forests of the eastern United States, reaching 80 to 100 feet in height at maturity and providing timber and low-fat, high-protein nutrition for animals and humans in the form of chestnuts. In the early 1900s an imported fungal pathogen, which causes chestnut leaf blight, was introduced into the United States. The disease spread from infected trees in the New York City area and by 1950 had killed more than 3 billion trees and virtually eliminated the chestnut tree in the United States. Economic losses in both timber and nut production have been estimated in the hundreds of billions of dollars. As well, there are ecological impacts in eliminating a dominant species.

To try to bring this tree back to the U.S. forests, several methods have been explored, including the development of blight-resistant varieties. We focus instead on attempts to weaken the fungus by infecting it with a virus that reduces the fungus' virulence. The basic idea is to release these hypovirulent isolates of chestnut blight fungus and let the viruses infect the natural populations of the fungus, thereby allowing chestnuts trees to survive.

Michael Milgroom from the Department of Plant Pathology at Cornell University, and his colleague, Paolo Cortesi from the University of Milan, have studied this system (Cortesi et al., 1996; Cortesi and Milgroom, 1998). Viruses can spread between fungal individuals only when they come in contact and fuse together. A major obstacle in spreading this virus and thus controlling the disease is that different isolates of the fungus cannot necessarily transfer the virus to one another. They have worked with six incompatibility genes, which may block the transmission of this virus between isolates of the fungus. By developing lab isolates that are compatible with a wide variety of naturally occurring isolates (and thus able to transmit the hypovirulence) an avenue may be opened to biocontrol of this fungal disease.

To estimate the effects of these genes Milgroom and Cortesi have made extensive attempts to pair isolates which differ on the first gene only, the second gene only, the first and the second gene, and so on. For each combination of isolates they attempt transmission an average of 30 times and record a binary response of whether or not the attempt succeeded in transmitting the virus.

Questions of interest include whether pre-identified genes actually do have an influence on transmission of the virus (and if so, to what

degree), whether there are other, as yet unidentified, genes that might affect transmission, and whether transmission is symmetric. By symmetry of transmission we mean the following: Suppose the infected fungus is type b at the locus for the first gene and the non-infected isolate (that we are trying to infect) is type B . The two isolates are the same at the other five loci. Is the probability of transmission the same as when using type B to try to infect type b ?

a. A random effects probit model

A common model in genetics for describing the presence or absence of a trait is the threshold model. This arises from assuming that a large number of genes each have a small and additive effect and that when the cumulative effect exceeds a threshold of zero, the trait is present in an individual. Letting $y = 1$ denote the presence of the trait, ϵ represent the additive genetic effect and $\mathbf{x}'\beta$ represent either genetic or non-genetic fixed effects, we can appeal to the central limit theorem to give the probit model:

$$\begin{aligned} P\{y = 1\} &= P\{\mathbf{x}'\beta + \epsilon > 0\} \\ \epsilon &\sim \mathcal{N}(0, 1), \end{aligned} \tag{8.48}$$

so that we have

$$P\{y = 1\} = P\{-\epsilon < \mathbf{x}'\beta\} = \Phi(\mathbf{x}'\beta)$$

or

$$\Phi^{-1}(\pi) = \mathbf{x}'\beta,$$

where $\pi = E[y] = P\{y = 1\}$.

We use this model by letting y_i be equal to 1 if the attempt succeeds in transmitting the virus and 0 otherwise.

– i. The fixed effects

We concentrate first on building the fixed effects portion of the model. With $\mathbf{x}'_i$ the i th row of the model matrix for the fixed effects, our model is

$$\mathbf{x}'_i\beta = \mu + \sum_{j=1}^6 \alpha_j MCH_j + \sum_{j=1}^6 \gamma_j ASY_j, \tag{8.49}$$

where MCH_j is 1 if there is a mismatch at locus j and zero otherwise and ASY_j is $1/2$ if there is a mismatch at locus j with a b donor, $-1/2$ if it is a B donor and 0 if there is no mismatch. The effect of a mismatch on gene j (averaged over donor types b and B) is thus measured by α_j , and γ_j measures the difference between a mismatch with a donor type b and a donor type B .

– ii. *The random effects*

The fact that different isolates of the fungus are used which may differ with regard to genes other than the six pre-identified suggests that we might model their effects as being selected from a normal distribution. Let $\mathbf{Z}_D$ be the model matrix for the donor effects, i.e., an incidence matrix identifying which donor isolates are used for which attempted transmissions. Similarly define $\mathbf{Z}_R$ for the recipient isolates, with the i th row of the matrices denoted respectively as $\mathbf{z}'_{iD}$ and $\mathbf{z}'_{iR}$. With $\mathbf{x}'_i$ defined by (8.49), $\mathbf{Z} = [\mathbf{Z}_D \ \mathbf{Z}_R]$, and $\mathbf{u}' = [\mathbf{u}'_D \ \mathbf{u}'_R]$, a reasonable model might then be:

$$\begin{aligned} P\{y_i = 1|\mathbf{u}\} &= \Phi(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_{iD}\mathbf{u}_D + \mathbf{z}'_{iR}\mathbf{u}_R) \\ &= \Phi(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u}) \\ \mathbf{u}_D &\sim \mathcal{N}(0, \mathbf{I}\sigma_D^2) \\ \mathbf{u}_R &\sim \mathcal{N}(0, \mathbf{I}\sigma_R^2) \\ \mathbf{u} &\sim \mathcal{N}(0, \mathbf{D}). \end{aligned} \tag{8.50}$$

In this model, $\mathbf{u}_D$ represents the (random) effects of the donor isolate and $\mathbf{u}_R$ represents the (random) effects of the recipient isolate.

– iii. *Consequences of having random effects*

The unconditional mean is given by

$$\begin{aligned} E[y_i] &= E[E[y_i|\mathbf{u}]] \\ &= E[\Phi(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u})]. \end{aligned}$$

This last quantity can most easily be calculated by appeal to the threshold model. We do not necessarily need to believe that the threshold model holds, but can merely use it as a probabilistic identity.

$$E[y_i] = E[P\{\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u} + \epsilon_i > 0|\mathbf{u}\}]$$

$$\begin{aligned}
&= P\{\mathbf{x}_i'\boldsymbol{\beta} + \mathbf{z}_i'\mathbf{u} + \epsilon_i > 0\} \\
&= P\{-(\mathbf{z}_i'\mathbf{u} + \epsilon_i) < \mathbf{x}_i'\boldsymbol{\beta}\} \\
&= P\{W < \mathbf{x}_i'\boldsymbol{\beta}\},
\end{aligned} \tag{8.51}$$

where $W \sim \mathcal{N}(0, \mathbf{z}_i'\mathbf{D}\mathbf{z}_i + 1)$. The marginal probability is thus

$$\begin{aligned}
E[y_i] &= \Phi\left(\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sqrt{\mathbf{z}_i'\mathbf{D}\mathbf{z}_i + 1}}\right) \\
&= \Phi(\mathbf{x}_i'\boldsymbol{\beta}^*),
\end{aligned} \tag{8.52}$$

where $\boldsymbol{\beta}^*$ is equal to $\boldsymbol{\beta}/\sqrt{\mathbf{z}_i'\mathbf{D}\mathbf{z}_i + 1}$.

This result is interesting in two ways. First, it is somewhat surprising (and, as it turns out, special) that the form of the relationship of the mean of $\mathbf{y}$ to the fixed effects is probit either conditionally or unconditionally. Second, it shows that the marginal coefficients on the probit scale are always *attenuated* as compared to the conditional coefficients. Thus it clearly is important to keep in mind when considering any of these models whether they represent the response conditional on the random effects or are, instead, marginal calculations.

Since y_i is binary, it has a marginal Bernoulli distribution with mean, $E[y_i]$, given by (8.52). Its variance is therefore $E[y_i](1 - E[y_i])$.

A typical consequence of including random effects is that they induce a correlation between observations sharing the random effects and this model is no exception. From first principles, the covariance of two observations with the same donor and recipient isolates would be given by $\text{cov}(y_i, y_j) = E[y_i y_j] - E[y_i]E[y_j]$. The second part of this can be evaluated using (8.52) and the first part calculated as

$$E[y_i y_j] = \int_{-\infty}^{\infty} \Phi(\mathbf{x}_i'\boldsymbol{\beta} + \sigma z) \Phi(\mathbf{x}_j'\boldsymbol{\beta} + \sigma z) \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz, \tag{8.53}$$

where $\sigma = \sqrt{\mathbf{z}_i'\mathbf{D}\mathbf{z}_j}$.

– iv. *Likelihood analysis*

Again, with a conditional specification, the likelihood is most naturally calculated by first writing out the conditional distribution and then integrating out the random factors. The conditional distribution given

$\mathbf{u}$ is the product of Bernoulli densities:

$$f_{\mathbf{Y}|\mathbf{u}}(\mathbf{y}|\mathbf{u}) = \prod_i \Phi(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u})^{y_i} [1 - \Phi(\mathbf{x}'_i\boldsymbol{\beta} + \mathbf{z}'_i\mathbf{u})]^{1-y_i}. \quad (8.54)$$

The likelihood would then be given by

$$L(\boldsymbol{\beta}, \mathbf{D}|\mathbf{y}) = \int f_{\mathbf{Y}|\mathbf{u}}(\mathbf{y}|\mathbf{u}) f_{\mathbf{U}}(\mathbf{u}) d\mathbf{u}, \quad (8.55)$$

where the integral is of order equal to the dimension of $\mathbf{u}$, which in this example is 259. Furthermore, for the design of this experiment, the likelihood does *not* break down into smaller-dimensional pieces, as it might with longitudinal data. This poses a serious computational problem.

– v. *Results*

Given the difficulty of calculating the likelihood, the techniques of Section 10.3c were used to fit the model. A logistic version of (8.54) was fitted using the Monte Carlo Newton–Raphson technique. This was done since the computations were somewhat faster than for (8.54). The maximized value of the likelihood was estimated by importance sampling.

The variance components were estimated to be

$$\hat{\sigma}_D^2 = 1.6$$

$$\hat{\sigma}_R^2 = 0.5$$

indicating a small to moderate correlation among observations taken on the same donor isolate and a somewhat smaller correlation among observations taken on the same recipient isolate. A likelihood ratio test of $H_0: \text{all } \gamma_i = 0$ gives a value for $-2 \log \Lambda$ of about 160 (since the value is determined by simulation it is not known exactly), which is highly statistically significant when referred to a chi-square distribution with 6 *df*. This indicates that, unfortunately, transmission is asymmetric: it depends on the value at that locus, not just on whether or not there is a match.

Further analysis shows that the fourth gene (tentatively identified from previous research) does not have an effect on transmission. Neither its direct effect nor its asymmetry effect is statistically significant.

8.9 EXERCISES

E 8.1 Show that, if $u_i \sim \mathcal{N}(0, \sigma_u^2)$ and that each row of $\mathbf{Z}$ has a single entry equal to 1 with all the rest being zero, then $M_u(\mathbf{z}) = \exp\{\sigma_u^2/2\}$.

E 8.2 Prove (8.12).

E 8.3 Show that the log likelihood for (8.15) can be written as (8.16).

E 8.4 Suppose that $Y = \delta X$, where $P\{\delta = 1\} = P\{\delta = 0\} = \frac{1}{2}$ and $X \sim \chi_1^2$ independent of δ . Show that

$$P\{Y > \chi_{1,1-2\alpha}^2\} = \alpha.$$

E 8.5 Derive the log likelihood for the Poisson-gamma model described in Section 8.6d.

E 8.6 Show that for (8.37), where the u_i are treated as fixed, unknown parameters to be estimated, that the MLE of β is given by (8.37). *Hints:* Consider separately pairs in which there are zero, one and two successes and first maximize with respect to the u_i , then β .

E 8.7 Show that $\hat{\beta}$ of (8.37) converges to 2β . *Hint:* Calculate $P\{y_{i1} = 1, y_{i2} = 0\}$ and $P\{y_{i1} = 0, y_{i2} = 1\}$ and hence the expected value of N_{01} and N_{10} .